Class: Final Year B. Tech. **Semester: VII**
Course Name: Research Ecosystem **Course Code: DJS23ITELX12**

Tutorial 1: Literature Review and Research Gap

Students are expected to submit a soft copy document and give a short presentation.

PART A – Cover Page

1. Proposed Research Title
The document should begin with the proposed research title followed by:

• Names of Students

• Roll Numbers

• Class/Division

• Academic Year: 2026–27

• Subject: Research Ecosystem

• Tutorial: 1 – Literature Review and Research Gap

PART B – Research Topic

1. Proposed Research Title
Write a specific and focused title.

Avoid broader title: E.g., Artificial Intelligence in Healthcare

Option1: 1. Machine Learning-Based Early Prediction of Cardiovascular Disease Using Clinical Data

2. Comparative Analysis of Ensemble Machine Learning Models for Early Cardiovascular Disease Prediction

Guidelines for writing a title: Title should include

• What is being studied?

• Which technology/method is being used?

• For what problem/application?

• If relevant, what type of data is being used?

2. Background and Problem Statement
Write approximately 1–2 pages explaining the research area.

Address:

• What is the broader problem?

• Why is the problem important?

• Who is affected by the problem?

• What solutions currently exist?

• Where is AI/ML/Data Science being used?

• Why is further research required?

Recommended structure:

Paragraph 1 – Introduce the domain.

Paragraph 2 – Explain the specific problem.

Paragraph 3 – Explain existing technological approaches.

Paragraph 4 – Explain limitations/challenges.

Paragraph 5 – Lead towards the research problem.

3. Problem Statement
The problem statement should be specific and researchable.

Students should answer: What problem are we trying to solve, and why are existing approaches insufficient?

Suggested format:

"Although _____ is widely used for _____, existing approaches suffer from _____. These limitations affect _____. Therefore, there is a need to investigate _____."

PART C – Literature Review

4. Literature Search Strategy
Mention how the literature was searched.

Recommended sources:

- Google Scholar
- IEEE Xplore

- ScienceDirect
- Springer
- ACM Digital Library
- PubMed (for healthcare-related topics)
- Scopus-indexed journals
- Web of Science Journals

Do not rely primarily on **random blogs, Wikipedia, YouTube, generic websites, or ChatGPT-generated references**.

5. Suggested Search Keywords
List the keywords and combinations used.

Example topic: ML-based disease prediction

• "machine learning" AND "disease prediction"

• "machine learning" AND "early disease prediction"

• "ensemble learning" AND "disease prediction"

• "XGBoost" AND "healthcare prediction"

• "explainable AI" AND "disease prediction"

Use multiple combinations rather than searching only the exact project title.

6. Literature Selection Criteria
Prefer papers that are:

• Recent

• Peer-reviewed

• Relevant to the research problem

• Closely related to the proposed methodology

• Based on credible datasets

• Published in reputed journals/conferences

Suggested literature: approximately 5-8 research papers, including at least 3–5 recent papers (preferably from the last 3 years) and 2–3 foundational/older papers where necessary. The exact number may be adjusted depending on the topic.

PART D – Literature Review Table

7. Literature Review Matrix

The literature review must compare papers rather than merely list them. Students should summarise each paper in their own words and identify its limitations/gaps.

What to Identify in Every Paper

For each paper, identify:

1. Research problem – What problem did the researchers address?

2. Dataset – Name, number of samples, features/modalities, source.

3. Methodology – Algorithms/models/approaches used.

4. Evaluation – Metrics used.

5. Results – Main findings.

6. Limitations – What was not addressed? This is particularly important for identifying the research gap.

Index for Literature

Sr. No	Title of the Paper	Scopus/WoS/IEEE	DOI
1			
2			
3			
4			
5			

Literature Review Matrix – Template

Sr. No.	Authors & Year	Research Problem	Dataset	Method / Model	Evaluation Metrics	Key Findings	Limitations / Gap
1							
2							
3							
4							
5							

PART E – Literature Synthesis

8. Literature Synthesis

After the table, write 1–2 pages of synthesis.

Answer: What does the literature collectively tell us?

Possible themes:

• Algorithms – Which approaches dominate?

• Dataset – What types and sizes of datasets are used?

• Evaluation – Which metrics are commonly reported?

• Explainability – Are model decisions explained?

• Generalization – Are models validated across datasets/populations?

• Real-world deployment – Has the method been evaluated under realistic conditions?

The synthesis should gradually lead to the research gap.

PART F – Research Gap Identification

9. Research Gap

Do not simply write: "There is no research on this topic."

Instead, identify a specific limitation or unexplored aspect.

Types of Research Gaps

Students can look for:

1. Methodological Gap – Existing studies use one approach while another has not been sufficiently investigated.

2. Dataset Gap – Existing research uses one dataset but has not been evaluated on another relevant dataset.

3. Performance Gap – Limitations in accuracy, recall, generalization or robustness.

4. Evaluation Gap – Previous studies evaluate only a limited set of metrics.

5. Explainability Gap – Models predict but do not explain decisions.

6. Generalization Gap – Good performance on one dataset but limited external validation.

7. Integration Gap – Different data modalities have been studied separately but not effectively combined.

8. Real-world/Deployment Gap – Good experimental performance but limited realistic evaluation.

9. Comparative Gap – A systematic comparison under the same experimental conditions is missing.

Suggested Gap-Writing Structure
"Existing research has demonstrated _____. However, most studies _____. Furthermore, _____ remains insufficiently investigated. Therefore, there is a need to _____."

PART G – Research Question

10. Research Question
The research question should directly emerge from the research gap.

Poor: How can AI be used in healthcare? (Too broad.)

Better: Which machine learning algorithm provides the best performance for cardiovascular disease prediction?

Stronger: How do ensemble learning methods compare with conventional machine learning methods for cardiovascular disease prediction under imbalanced-class conditions?

Research Question Checklist
Before finalizing, check:

• Is it specific?

• Is it researchable?

• Can it be answered using data/experiments?

• Is it connected to the research gap?

• Can it realistically be completed within the project duration?

• Does it have measurable outcomes?

PART H – Research Objectives

11. Research Objectives
Objectives should explain what the students will actually do.

Good action verbs:

Identify, Analyze, Develop, Implement, Compare, Evaluate, Investigate, Assess, Validate, Optimize.

Avoid vague verbs such as:

Understand, Study, Know, Learn.

Example:

1. To identify relevant clinical features for disease prediction.

2. To preprocess and analyze the selected dataset.

3. To develop baseline machine learning models.

4. To implement ensemble learning models.

5. To compare models using accuracy, precision, recall, F1-score and ROC-AUC.

6. To analyze the contribution of important features to model predictions.

PART I – Scope of Research

12. Scope

Clearly define what the project will and will not cover.

Include:

• Data Scope – What data will be used?

• Population Scope – Who/what does the data represent?

• Technical Scope – Which algorithms/technologies will be considered?

• Evaluation Scope – Which metrics will be used?

• Application Scope – Where can the proposed solution be used?

Example:

"The study will focus on structured clinical data for cardiovascular disease prediction. The study will evaluate Logistic Regression, Random Forest and XGBoost models. Performance will be assessed using F1-score, recall, precision and ROC-AUC. The study will focus on model development and evaluation and will not include clinical deployment or real-time diagnosis."

PART J – Research Agenda

13. Research Agenda

The research agenda is the roadmap for the project.

Suggested phases:

Phase 1 – Literature Review → Literature matrix

Phase 2 – Gap Identification → Research gap

Phase 3 – Dataset Collection → Final dataset

Phase 4 – Data Preprocessing → Clean dataset

Phase 5 – Baseline Model → Baseline results

Phase 6 – Proposed Model → Model/results

Phase 7 – Evaluation → Comparison

Phase 8 – Analysis → Findings

Phase 9 – Documentation → Final report

PART K – References

14. References

Use a consistent referencing style, preferably IEEE.

Example:

[1] A. Author and B. Author, "Title of paper," Journal Name, vol. X, no. Y, pp. XX–XX, 2024.

Important:

• Every paper in the literature table should appear in the reference list.

• Every paper cited in the document should have a corresponding reference.

• Do not include references that were not actually consulted.

PART L – Soft Copy Submission

15. Mandatory Document Contents

Submit one PDF containing:

1. Cover Page

2. Research Title

3. Background

4. Problem Statement

5. Literature Search Strategy

6. Literature Review Table

7. Literature Synthesis

8. Research Gap

9. Research Question

10. Research Objectives

11. Scope

12. Research Agenda

13. References

Also submit soft copies of the selected research papers or a properly organized folder containing the papers.

PART M – Uploading on teams

16. Three PDFs
1. IPD_Tutorial_1_Name_Roll No.pdf

2. Research Papers_Name_Roll No.pdf

3. Presentation.pptx

PART N – Presentation Guidelines

17. Presentation Structure
Keep the presentation to 6–8 slides.

Slide 1 – Title and team members

Slide 2 – Problem and motivation

Slide 3 – Literature review

Slide 4 – Comparison of literature

Slide 5 – Research gap

Slide 6 – Research question and objectives

Slide 7 – Scope

Slide 8 – Research agenda

18. Presentation Focus
The Research Gap slide is the most important. Clearly show:

Existing work → Limitation → Gap → Proposed direction